Math. 2 ①

Indefinite Integral

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Def: let $f(x)$ & $F(x)$ be two function defined on $]a, b[$ and

$$\frac{dF(x)}{dx} = f(x) \quad ; \quad \forall x \in]a, b[$$

Then $F(x)$ is said to anti-derivative of $f(x)$ denoted by

$$\boxed{F(x) = \int f(x)\, dx}$$

Note: If $F(x)$ anti-derivative of $f(x)$ then $F(x) + c$ is also anti-derivative of $f(x)$ because $[F(x)+c]' = F(x)$

اي ان الدالة المقابلة لـ $f(x)$ ليست وحيدة

(i.e) $$\boxed{\int f(x)\, dx = F(x) + \underset{\text{Constant of integration}}{c}}$$

Stander Rules of Indefinite Integral:

(1) $\int K f(x)\,dx = K \int f(x)\,dx$

(2) $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

(3) $\int x^r\,dx = \frac{x^{r+1}}{r+1} + c$

(4) $\int [f(x)]^n \frac{df}{dx}\,dx = \frac{[f(x)]^{n+1}}{n+1} + c$

(5) $\int e^{f(x)} f'(x)\,dx = e^{f(x)} + c$

(6) $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$

(7) $\int a^{f(x)} \frac{df}{dx}.dx = \frac{a^{f(x)}}{\ln a} + c$

Integration of Trigonometric functions:

(1) $\int \sin f(x) . \frac{df}{dx}\,dx = -\cos f(x) + c$

(2) $\int \cos f(x) . \frac{df}{dx}\,dx = \sin f(x) + c$

(3) $\int \sec^2 f(x) . \frac{df}{dx}\,dx = \tan f(x) + c$

(4) $\int \operatorname{cosec}^2 f(x) \frac{df}{dx}\,dx = -\cot f(x) + c$

(5) $\int \sec f(x) \tan f(x) \frac{df}{dx}\,dx = \sec f(x) + c$

(6) $\int \operatorname{cosec} f(x) \cot f(x) \frac{df}{dx}\,dx = -\operatorname{cosec} f(x) + c$

Integration of Hyperbolic functions :

1. $\int \sinh x\,dx = \cosh x + c$
2. $\int \cosh x\,dx = \sinh x + c$
3. $\int \operatorname{sech}^2 x\,dx = \tanh x + c$
4. $\int \operatorname{csch}^2 x\,dx = -\coth x + c$
5. $\int \tanh x \operatorname{sech} x\,dx = -\operatorname{sech} x + c$
6. $\int \coth x \operatorname{csch} x\,dx = \operatorname{csch} x + c$

* Integrations gine invers trigonometric func.

1. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + c$
2. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$
3. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \frac{x}{a} + c$

كل العلاقات السابقة تسمى علاقات قياسية
لأنها تنتج من التفاضل مباشرة

Example: Find the following integrals

(1) $\int \frac{x+1}{\sqrt{x}}\,dx$

(2) $\int \frac{\sin x}{\cos^2 x}\,dx$

(3) $\int \left(\frac{1}{2\sqrt{x}} - 3\cos x + 3\sec x \tan x\right) dx$

(4) $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} + \cos x + \sec^2 x\right) dx$

(5) $\int \left(\sqrt{x} - \frac{1}{x}\right)^2 dx$

(6) $\int (x^2+1)^2 (x)\,dx$

(7) $\int \sin^3 x \cos x\,dx$

(8) $\int \sin(3x+8)\,dx$

(9) $\int \sin^7 3x \cos 3x\,dx$

(10) $\int \sqrt{x^2+1}\,(2x)\,dx$

(11) $\int \frac{\cos 2x}{\sqrt{1+\sin 2x}}\,dx$

(12) $\int \frac{2x}{\sqrt{2x-1}}\,dx$

(13) $\int \frac{\cos\sqrt{x}}{\sqrt{x}}\,dx$

(14) $\int x \sec(x^2) \tan(x^2)\,dx$

(15) $\int \frac{x^2+1}{x^3+3x}\,dx$

(16) $\int \frac{\sqrt{\ln x}}{x}\,dx$

(17) $\int \sec x\,dx$

(18) $\int \csc x\,dx$

(19) $\int \tan x\,dx$

(20) $\int \cot x\,dx$

(21) $\int \sin^2 x \, dx$

(22) $\int \cos^2 x \, dx$

(23) $\int \tan^2 x \, dx$

(24) $\int \cot^2 x \, dx$

(25) $\int \sin 5x \cos 7x \, dx$

~~(26) ∫~~

(26) $\int \frac{x^2+x+1}{x^2+1} \, dx$

(27) $\int \frac{dx}{x \ln x}$

(28) $\int e^{3x+1} \, dx$

(29) $\int \frac{e^{3/x}}{x^2} \, dx$

(30) $\int 5^{\cos x} \sin x \, dx$

(31) $\int \frac{x+1}{x^2+2x+5} \, dx$

(32) $\int \frac{\tan x - \sec x}{\cos x} \, dx$

(33) $\int \frac{dx}{1+3x^2}$

(34) $\cancel{\int \frac{e^{2x}}{\sqrt{1-e^{4x}}} \, dx}$

$\cos 2x = 1 - 2\sin^2 x$

$= 2\cos^2 x - 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \csc^2 x$

علاقات مهمة

$\sin x \cos y = \frac{1}{2}\left[\sin(x-y) + \sin(x+y)\right]$

$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right]$

$\cos x \cos y = \frac{1}{2}\left[\cos(x-y) + \cos(x+y)\right]$